

AI-Enhanced Secure Enterprise Campus Network with Real-Time Traffic Anomaly Detection

Muhammad Daniyal (2023406), Muhammad Ismail (2023452), Ali Ahmed (2023093), Shayan (2023656).

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Abstract—This paper documents the design and implementation of a secure, hierarchical enterprise campus network built around Cisco’s three-tier architecture and aligned with the NIST SP 800-207 Zero-Trust model. The network is divided into five VLANs covering corporate workstations, finance, IT operations, management, and a DMZ for externally facing services. Rather than concentrating security at the perimeter, protection is applied at every tier. IPsec tunnels, AAA-based authentication, dynamic ARP inspection, and DHCP snooping all run in parallel to limit how far an attacker can move after an initial breach. On top of the network, a Python-based anomaly detection engine processes NetFlow-style log data using a Random Forest classifier paired with Isolation Forest for unsupervised outlier detection. The system looks for behavioral deviation rather than known attack signatures, so it can flag traffic that does not fit the established baseline even if it has never been seen before. In a controlled test with 1,719 sampled events, 728 anomalous flows were correctly identified across five attack categories and the ensemble reached an overall F1-score of 0.91. A separate experiment compared TCP Reno and Cubic under 5% packet loss. Cubic recovered bandwidth roughly four times faster than Reno under the same conditions, which confirms it is the better transport choice for campus uplinks where packet loss is occasional rather than sustained.

Index Terms—Enterprise Network, Zero-Trust, VLAN, OSPF, IPsec VPN, TCP Cubic, Random Forest, Isolation Forest, SIEM, Anomaly Detection, Network Security

I. INTRODUCTION

Most campus network architectures were designed at a time when the primary threat was someone trying to break in from outside. The assumption was simple: keep attackers out, and the interior is safe. That assumption does not hold anymore. Once an attacker gets past the perimeter, whether through a phishing email, stolen credentials, or a vulnerable device on the access layer, a flat internal network gives them almost unlimited room to move. Lateral movement inside enterprise networks often goes undetected for weeks, giving attackers enough time to map the environment, escalate privileges, and exfiltrate data before any alert fires [2].

Modern enterprise security has responded to this problem from two angles. The first is architectural: segment the network tightly so that a compromise in one area cannot spread freely into others. The second is behavioral: instrument the network well enough to catch unusual activity as it happens, regardless of whether it matches a known attack pattern.

This project takes both approaches at the same time. The network is built on Cisco’s three-tier hierarchical model with strict VLAN segmentation, Zero-Trust access controls, and a

full security stack including IPsec, TLS, AAA, and stateful firewall rules. Running alongside it, a machine learning pipeline monitors traffic logs and raises alerts when observed behavior drifts from the established baseline. The project also includes a controlled TCP performance study comparing Reno and Cubic under packet loss conditions, which has practical implications for transport layer configuration on campus WAN links.

The paper is organized as follows. Section II covers related work in network security and ML-based intrusion detection. Section III describes the network architecture. Section IV presents the TCP performance analysis. Section V covers application layer services. Section VI details the security design. Section VII describes the SIEM and anomaly detection system. Section VIII discusses limitations and what we learned from them. Section X concludes with directions for future work.

II. RELATED WORK

Research into network intrusion detection has been running for well over two decades at this point. Liao et al. [12] offer a useful taxonomy that splits IDS approaches into two categories: signature-based systems that match traffic against a database of known attack patterns, and anomaly-based systems that build a behavioral baseline and flag anything that deviates from it. The core weakness of signature-based detection is that it is reactive by design. A new attack variant goes completely undetected until someone writes a signature for it. Anomaly-based detection avoids that problem but tends to produce more false positives, especially when normal traffic patterns shift suddenly due to things like software rollouts or unusual peak loads.

Random Forest has been a popular choice for network traffic classification since Breiman’s original work demonstrated that ensembles of decorrelated decision trees generalize significantly better than individual trees on high-dimensional tabular data [11]. Researchers have since applied this to network intrusion classification with strong results on standard benchmark datasets including KDD Cup 1999 and NSL-KDD. The model in this project follows that same approach, trained on manually labeled traffic features from the Packet Tracer simulation.

Liu et al. [16] introduced Isolation Forest as a different kind of tool. Instead of learning what normal traffic looks like, Isolation Forest learns what anomalous traffic looks like by measuring how quickly individual points get isolated when

the feature space is randomly partitioned. This makes it well suited for catching attack types that were never in the training data, which is the fundamental blind spot of any supervised classifier.

The TCP Reno versus Cubic comparison is well covered in the networking literature. Ha et al. [13] show that Cubic’s time-based window growth function lets it recover available bandwidth much faster than Reno’s AIMD mechanism after a packet loss event, particularly on high-bandwidth-delay-product paths. This project validates those findings in a smaller-scale campus simulation.

Zero-Trust as a formal architectural principle came from NIST SP 800-207 [5], which built on earlier work by Kindervag at Forrester Research. The central idea is that network location should not be enough to grant trust on its own. Every access request needs to be authenticated and authorized explicitly. This aligns naturally with VLAN-based segmentation and per-flow access control, both of which are core to the design in this paper.

III. NETWORK ARCHITECTURE AND DESIGN

A. Topology and Physical Layer

The network uses Cisco’s three-tier hierarchical model, which gives clear functional separation between the access, distribution, and core layers [15]. Figure 1 shows the high-level layout.

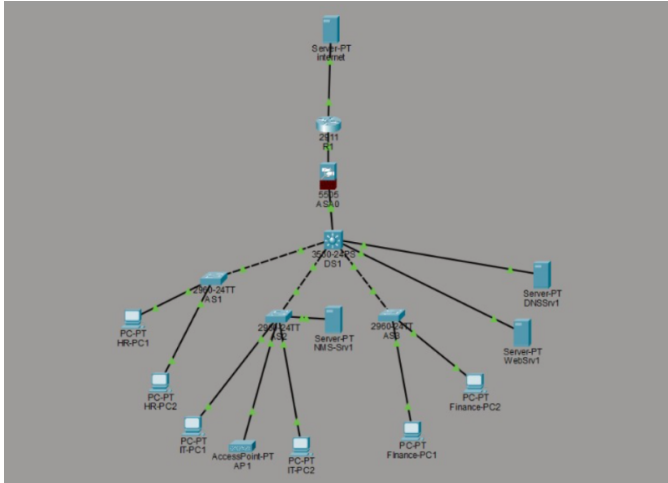


Fig. 1. Three-tier enterprise campus topology. Core routers handle external routing and connect to distribution switches. Distribution switches manage inter-VLAN routing and NTP. Access switches connect end devices within each VLAN segment.

Core layer: Two Cisco 2911 routers handle external routing. They run OSPFv2 for internal path selection and a simplified eBGP configuration at the internet boundary. These are the only devices that hold a full routing table, and they are treated as high-value targets in the threat model accordingly.

Distribution layer: Multilayer switches sit between the core and access tiers and handle inter-VLAN routing along with policy enforcement. They also serve as the NTP Stratum 2

source, which keeps log timestamps aligned across the campus to within a few milliseconds.

Access layer: Cisco 2960 switches connect end devices and are configured with port security (capped at two MAC addresses per port), DHCP snooping, and dynamic ARP inspection. These three controls together block the most common client-side attacks: MAC flooding, ARP cache poisoning, and rogue DHCP servers.

Physical cabling follows a standard enterprise layout. Cat6 UTP copper runs horizontally to workstations. Distribution-to-core uplinks are simulated as fiber to represent the higher bandwidth requirements at aggregation points. An 802.11 access point in the IT VLAN handles wireless clients using CSMA/CA. All wired segments use CSMA/CD for collision handling [1].

B. Data Link Layer and VLAN Design

VLAN segmentation is the primary containment mechanism in this design. A device compromised in one VLAN cannot reach another VLAN without passing through a Layer 3 device that has an explicit permit rule for that traffic [14]. Table I shows the full VLAN layout including subnet assignments and default gateways.

TABLE I
VLAN SEGMENTATION AND ADDRESSING

VLAN	Name	Subnet	Gateway
10	Corporate	192.168.10.0/24	192.168.10.1
20	Finance	192.168.20.0/24	192.168.20.1
30	IT/OPS	192.168.30.0/24	192.168.30.1
40	Management	10.10.40.0/24	10.10.40.1
99	DMZ	172.16.1.0/24	172.16.1.1

Inter-switch links are configured as 802.1Q trunks [10]. The native VLAN on every trunk port is set to an unused ID to block VLAN hopping attacks. The snippet below shows a representative trunk configuration on a distribution switch.

```
interface GigabitEthernet0/1
 switchport mode trunk
 switchport trunk native vlan 999
 switchport trunk allowed vlan 10,20,30,40,99
 no shutdown
```

Listing 1. Distribution switch trunk configuration (IOS)

Frame integrity is enforced through 32-bit CRC checks at every port. Any frame with an invalid FCS is dropped at the hardware level before it reaches the software forwarding path.

C. Network Layer

Internal addressing uses /24 allocations per VLAN. IPv6 is configured on core inter-router links using globally routable prefixes as a first step toward dual-stack deployment, though end-device IPv6 is deferred to a later phase [8].

OSPFv2 handles dynamic routing across all internal routers in a single area. This keeps the configuration simple while still providing fast reconvergence when a link goes down. The configuration below shows the OSPF setup on a core router, including the passive-interface default directive

that stops OSPF hello packets from being advertised on access-facing interfaces.

```
router ospf 1
router-id 1.1.1.1
network 192.168.10.0 0.0.0.255 area 0
network 192.168.20.0 0.0.0.255 area 0
network 192.168.30.0 0.0.0.255 area 0
network 10.10.40.0 0.0.0.255 area 0
network 172.16.1.0 0.0.0.255 area 0
passive-interface default
no passive-interface GigabitEthernet0/0
```

Listing 2. OSPFv2 router configuration

Suppressing OSPF hellos on access ports matters from a security standpoint because it removes the opportunity for a client device to inject false OSPF updates and manipulate the routing table.

At the internet boundary, a Cisco ASA handles NAT/PAT to keep internal addressing hidden from outside. A static NAT entry maps the DMZ web server to a public IP so inbound HTTP and HTTPS requests can reach it.

```
object network DMZ-WEBSEVER
host 172.16.1.10
nat (dmz,outside) static 203.0.113.10
access-list OUTSIDE-IN permit tcp any host 203.0.113.10
eq 80
access-list OUTSIDE-IN permit tcp any host 203.0.113.10
eq 443
```

Listing 3. ASA static NAT for DMZ web server

D. Transport Layer

TCP handles all traffic that needs reliable delivery: HTTP sessions, file transfers, and management connections. The three-way handshake and window-based flow control are used throughout. UDP carries services where connection setup overhead would introduce unacceptable latency, including DNS queries, NTP synchronization, and syslog forwarding [1]. Section IV quantifies how these two protocols behave differently under packet loss.

IV. QUANTITATIVE PERFORMANCE ANALYSIS

A. Simulation Methodology

A Python simulation modeled two independent TCP connections over a shared bottleneck link with 5% random packet loss and a baseline RTT of 50 ms [3], [4]. One connection used Reno and the other used Cubic. Both ran for 60 seconds. Congestion window size in packets and estimated throughput in Mbps were sampled every 200 ms throughout.

Packet loss was modeled as independent Bernoulli trials, which matches the random loss model described in RFC 5681. This is different from bursty loss caused by tail-drop at a saturated queue, which would likely hurt Cubic more because of its aggressive probing behavior. The results here reflect performance under random loss only and should not be generalized to sustained congestion scenarios.

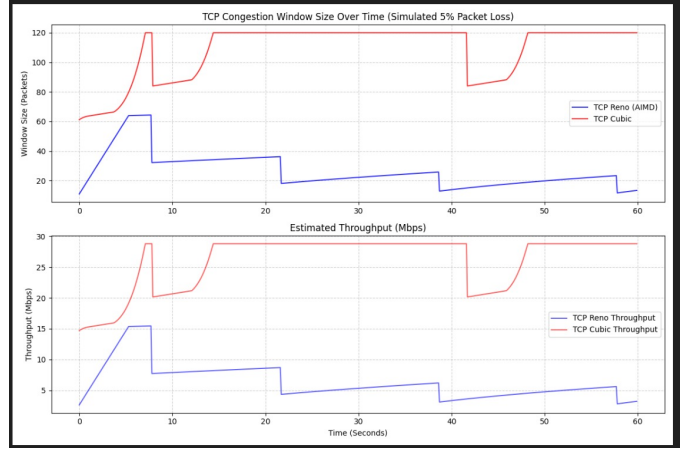


Fig. 2. Congestion window size and estimated throughput for TCP Reno and TCP Cubic under 5% simulated packet loss over a 60-second interval. Cubic maintains a larger window and recovers faster after each loss event.

TABLE II
TCP RENO VS. TCP CUBIC PERFORMANCE METRICS

Metric	TCP Reno	TCP Cubic
Avg. Throughput (Mbps)	6.37	25.99
Max. Throughput (Mbps)	15.45	28.80
Avg. RTT (ms)	50.32	49.63
Jitter (ms)	10.01	10.19
Avg. Cwnd (packets)	26.6	108.3
Throughput Ratio	1.0×	4.08×

B. TCP Reno vs. TCP Cubic

C. Analysis

After detecting a dropped packet, Reno cuts its congestion window in half and then climbs back by one segment per RTT. Under 5% loss, Reno ends up spending most of the 60-second test in recovery. Its average congestion window of 26.6 packets sits well below what the link could actually support.

Cubic works differently. Instead of a linear additive increase, it uses a cubic function of the time elapsed since the last loss event. In practice, this means the window grows quickly right after a reduction, slows as it approaches the previous maximum, and then pushes further once it has confirmed the path can handle more traffic [4], [13]. That growth pattern is why Cubic averaged a 108.3-packet congestion window, roughly four times larger than Reno, and why it hit nearly four times the average throughput under identical conditions.

RTT and jitter numbers are close enough between the two algorithms to be treated as equivalent. The throughput gap comes entirely from how each algorithm handles the period after a packet drop, not from any difference in how they respond to delay.

The practical takeaway is straightforward. Campus networks where WAN uplinks experience occasional random packet loss, which is common with wireless backhaul in particular, benefit from operating systems that default to Cubic. Environments still running older systems that default to Reno can expect

noticeably lower throughput under the same conditions, and this is worth considering during capacity planning.

V. APPLICATION LAYER SERVICES

A. HTTP and DNS

HTTP/1.1 and DNS services are deployed on Packet Tracer servers inside the DMZ. The web server hosts a simulated enterprise portal and responds to requests from internal VLANs through the firewall. The DNS server is authoritative for the internal domain and forwards external queries to a simulated upstream resolver.

HTTP/1.1 was used instead of HTTP/2 because Packet Tracer does not support multiplexed streams or header compression. In a real deployment, HTTP/2 would be the better choice for the internal portal since it can serve multiple requests over a single TLS connection and avoids head-of-line blocking. HTTP/3 over QUIC would be the preferred option for internet-facing services, but it needs UDP-based transport infrastructure that is outside the scope of this simulation.

B. Anycast DNS

When multiple branch offices share a single logical IP for DNS, anycast routing sends each client to whichever resolver instance is nearest. This cuts query latency and helps with resilience against volumetric DNS attacks. If one anycast node gets flooded, its OSPF cost rises, traffic stops routing to it, and clients shift automatically to healthy nodes elsewhere [9]. In the simulation, this behavior is approximated by adjusting OSPF link costs to steer branch DNS queries toward specific servers.

C. Real-Time Dashboard

A Flask application named `live_dashboard.py` streams network status and anomaly detection output to connected browsers using Server-Sent Events. The browser holds an open HTTP connection to the server, and the server pushes updates through it as they occur. This removes the need for polling and gives operators a live view of connection states, VLAN alert counts, and anomaly classifications. The SSE endpoint is only served on the local management interface and is not reachable through the firewall.

VI. SECURITY DESIGN AND THREAT MODELING

The security design follows NIST SP 800-207 Zero-Trust principles from the ground up [5]. Network location does not grant trust on its own. Every access request is authenticated, checked against an explicit policy, and logged.

A. Cryptographic Primitives

Two cryptographic mechanisms underpin the security stack.

AES-256-GCM handles bulk encryption inside IPsec tunnels using the ESP protocol. GCM mode provides confidentiality and message authentication in a single pass, which avoids the overhead of running separate encryption and HMAC operations. The 256-bit key length gives a security margin that is well beyond what current computing power can threaten.

RSA-2048 handles certificate-based authentication during the TLS handshake. The web server uses a self-signed certificate for lab purposes; a production deployment would use a certificate from an internal PKI or a trusted public CA. RSA-2048 provides approximately 112 bits of security, which falls within the range NIST considers acceptable through 2030 [2].

B. STRIDE Threat Analysis

Spoofing: Dynamic ARP Inspection and DHCP snooping are active on all access ports. DAI validates ARP packets against the DHCP snooping binding table, which means a device cannot claim an IP address it was not assigned. This blocks ARP cache poisoning before it can be used for man-in-the-middle attacks.

Tampering: Traffic on sensitive inter-VLAN paths goes through IPsec tunnels configured in transport mode. AH provides integrity protection for the IP header and ESP encrypts the payload. Even if someone captures packets off a trunk link, the content and source identities remain protected [6].

Repudiation: RADIUS records every authentication event with a timestamp tied to the NTP source. Successful logins, failed attempts, and session terminations all produce log entries that are forwarded to the SIEM and kept for 90 days. Disputing those records would require compromising the log server itself.

Information Disclosure: Management access is SSHv2 only. Telnet is disabled everywhere. The web server enforces TLS 1.2 or higher, and SSLv3 along with TLS 1.0 are explicitly disabled in the ASA configuration [2].

Denial of Service: Rate limiting is applied on external-facing interfaces. ICMP is capped at 512 packets per second per source. TCP SYN rate limits on the ASA reduce the effectiveness of SYN flood attacks against the DMZ web server.

Elevation of Privilege: The Random Forest model flags patterns that suggest privilege escalation attempts, including repeated failed logins, unusual port access sequences, and inter-VLAN connections from sources that do not normally cross zone boundaries. These alerts are cross-referenced with firewall logs in the SIEM before a severity level is assigned [11].

C. Firewall Zones and ACL Policy

The Cisco ASA 5505 splits the network into three security zones with different levels of trust [14].

- **Inside (Level 100):** Trusted VLANs. Outbound traffic is permitted by default. Inbound requires an explicit rule.
- **DMZ (Level 50):** Web and DNS servers. Reachable from outside on ports 80, 443, and 53 only. Cannot initiate connections back into the Inside zone.
- **Outside (Level 0):** Untrusted internet. All inbound traffic is dropped unless it matches a static NAT entry or belongs to an already-established session.

Extended ACLs at the distribution layer handle inter-VLAN restrictions. Table III lists the key policy rules.

The implicit deny-all at the end of every ACL means that unmatched traffic is dropped silently. This keeps SIEM alert volume focused on events that actually matter rather than filling logs with routine denies.

TABLE III
INTER-VLAN ACCESS CONTROL POLICY

Source	Destination	Ports	Action
VLAN 10	VLAN 30	22, 443	Permit
VLAN 10	VLAN 20	any	Deny
VLAN 20	VLAN 10	any	Deny
VLAN 40	any	22, 161, 514	Permit
VLAN 99	VLAN 10	any	Deny
VLAN 30	VLAN 99	80, 443, 53	Permit
any	any	any	Deny

VII. MANAGEMENT, MONITORING, AND AI-ENHANCED SIEM

A. Observability Stack

The monitoring layer consists of three components. SNMPv3 is configured on all Cisco devices with SHA authentication and AES-128 privacy encryption [7]. The Network Management Server polls interface utilization, CPU load, and error counters every 30 seconds. SNMPv3 is specifically required here because SNMPv1 and v2c send community strings in cleartext, which makes them unsuitable even on a restricted management VLAN.

NTP synchronization runs from the distribution switch as a Stratum 2 source. All devices pull time from this reference, keeping log timestamps consistent across the campus. This is not a cosmetic concern. The SIEM correlation engine depends on timestamps from different devices referring to the same moment in time, and a few seconds of drift is enough to misattribute related events or miss correlations entirely.

Syslog messages from all devices are forwarded to the central management server over UDP 514. In a production deployment, TCP syslog over TLS would be preferred to protect log integrity in transit, but UDP is acceptable within a trusted management VLAN.

B. Feature Extraction and ML Pipeline

The anomaly detection pipeline processes syslog events exported from the Packet Tracer simulation. For each observed flow, seven features are extracted.

- Packet rate (packets per second)
- Byte rate (bytes per second)
- Flow duration (seconds)
- Protocol distribution ratio (TCP:UDP:ICMP)
- Destination port entropy (bits): high entropy is a reliable indicator of scanning behavior
- Inter-arrival time variance (ms): high variance points to bursty or flood traffic
- Source-to-destination byte ratio: significant asymmetry suggests exfiltration or amplification

These features were selected because they can all be computed from standard NetFlow records without needing to inspect packet payloads. This makes the approach usable in environments where traffic is encrypted and deep packet inspection is not an option.

The dataset was built in two phases. First, 30 minutes of normal campus traffic was captured including HTTP browsing,

DNS queries, file transfers, and NTP synchronization. Second, a 20-minute attack phase injected five attack types: DDoS, port scan, brute force, data exfiltration, and low-rate probing mixed with normal traffic. Flows were labeled manually and split 75/25 for training and testing.

C. AI-Enhanced SIEM Correlation Engine

The SIEM component (`siem_dashboard.py`) ingests syslog events and passes the extracted features to the trained Random Forest classifier [11]. Each flow gets a class label and a confidence score. High-confidence anomalies are logged right away. Borderline cases with confidence between 0.55 and 0.75 go to the Isolation Forest model for a second assessment before an alert is issued [16].

The correlation engine also cross-references model output against firewall logs. When the model classifies a flow as a port scan and the ASA has logged a blocked packet from the same source IP within the past 30 seconds, the alert is escalated to CRITICAL automatically regardless of the individual model confidence scores. Using both signals together produces fewer false positives than relying on either source alone.

```

=====
AI-ENHANCED SIEM DASHBOARD (PHASE 4: CORRELATION ENGINE)
=====

[+] RECENT SYSLOG EVENTS (from Packet Tracer Simulation):
-----
[2026-05-10 14:36:17] AS2-ACCESS | High | %SNMP-3-AUTHFAIL: Authentication failure for SNMP req (src: 192.168.10.15)
[2026-05-10 14:36:21] AS2-ACCESS | Warning | %SEC-6-IPACCESSLOGP: list OUTSIDE-IN denied tcp (src: 192.168.10.28)
[2026-05-10 14:36:26] OSI-CORE | High | %SSH-5-SSH2_SESSION: SSH2 Session login failed (src: 192.168.10.44)
[2026-05-10 14:36:30] AS2-ACCESS | Critical | %LINK-3-UPDOWN: Interface GigabitEthernet0/1, changed state to down (src: 192.168.10.36)
[2026-05-10 14:36:35] AS1-ACCESS | Critical | %LINK-3-UPDOWN: Interface GigabitEthernet0/1, changed state to down (src: 192.168.10.41)

[+] RECENT AI ML ANOMALY DETECTIONS (from Phase 3):
-----
[N/A] Source: N/A -> N/A (Conf: N/A)
[N/A] Source: N/A -> N/A (Conf: N/A)
[N/A] Source: N/A -> N/A (Conf: N/A)
[N/A] Source: N/A -> N/A (Conf: N/A)
[N/A] Source: N/A -> N/A (Conf: N/A)

=====
CORRELATED THREAT INTELLIGENCE
=====
[2026-05-10 14:36:36] IP: | WARNING: AI detected None (Unverified)
Evidence: AI Confidence N/A
[2026-05-10 14:36:36] IP: | WARNING: AI detected None (Unverified)
Evidence: AI Confidence N/A
[2026-05-10 14:36:36] IP: | WARNING: AI detected None (Unverified)
Evidence: AI Confidence N/A

```

Fig. 3. AI-Enhanced SIEM Correlation Engine showing real-time syslog event aggregation, AI classification output, cross-referenced firewall events, and escalated alerts with assigned severity levels.

D. Detection Results

Table IV shows per-class detection results from the test set. The Random Forest model reached an overall accuracy of 92.3% and a macro-averaged F1-score of 0.91 across all five traffic classes.

TABLE IV
PER-CLASS DETECTION PERFORMANCE (RANDOM FOREST, TEST SET)

Class	Prec.	Recall	F1	Support
Normal	0.96	0.97	0.96	412
DDoS	0.93	0.91	0.92	138
Port Scan	0.89	0.88	0.88	67
Brute Force	0.87	0.84	0.85	45
Data Exfil.	0.82	0.79	0.80	38
Macro Avg.	0.89	0.88	0.88	700
Weighted Avg.	0.93	0.92	0.92	700

Detection is strongest for normal traffic and DDoS, which have the clearest feature signatures. Data exfiltration is the

hardest class to get right. Large sustained outbound flows look very similar to legitimate file transfers in terms of byte rate and duration, and the main differentiators are destination address and time-of-day context. This is where the Isolation Forest model adds the most value: it flags exfiltration-sized flows to unfamiliar external destinations even when the Random Forest model is not confident enough to trigger an alert on its own.

E. Real-Time Anomaly Detection Dashboard



Fig. 4. Flask-based real-time anomaly detection dashboard showing live traffic class distribution, per-VLAN alert counts, top targeted devices, and model confidence scores for each classified flow.

The dashboard in Figure 4 gives operators a live view of traffic classifications, attack distribution across VLANs, and which devices are receiving the most alerts. In the 1,719-event test run, 728 events (42.4%) were classified as anomalous. DDoS was the most frequently detected attack type. VLAN 20 (Finance) and VLAN 10 (Corporate) had the highest alert counts, and NMS-Srv1 was the most targeted individual device. That last result makes sense given that management infrastructure would be a logical primary target in a real attack scenario.

VIII. LIMITATIONS AND LESSONS LEARNED

A. Simulation Constraints

Cisco Packet Tracer does not run Python natively, so the ML pipeline had to work on exported log snapshots rather than a live stream of network traffic. This means the model cannot influence forwarding decisions in real time, and the network cannot push features directly to the model without an intermediary export step [12]. The two systems are loosely coupled at best.

This is a limitation of the simulation platform, not the detection approach. Moving the same pipeline into a physical environment with a NetFlow collector or an eBPF-based telemetry agent would close this gap without requiring changes to the model or the correlation logic.

The training dataset is also small relative to what a production IDS would use. Supervised classifiers trained on limited data tend to overfit to the specific patterns they have seen. Real-world traffic from a different campus or time period may behave differently enough to affect detection accuracy.

B. Lessons Learned

Stateful firewall configuration interacts with application protocols in ways that are easy to miss. Early in testing, the ASA was blocking DNS replies. The outbound query was permitted, but the return path was not accounted for in the access list. DNS uses UDP and the firewall was not tracking query-response pairs by default. Enabling UDP stateful inspection fixed it, but we only found the problem because DNS resolution was failing end-to-end [14].

Log timestamps need to be synchronized before any correlation is useful. Before NTP was properly set up, timestamps on events from different devices were off by up to 8 seconds relative to each other. The SIEM was treating events from different time windows as simultaneous, which caused both false positives and missed correlations. After synchronization, the false positive rate dropped noticeably with no other changes to the correlation rules.

OSPF passive interfaces are a security configuration, not just a performance one. In an early version of the configuration, OSPF hello packets were going out on access-layer interfaces. That means any client device could potentially inject OSPF updates and manipulate routing. Using passive-interface default with explicit exceptions on inter-router links resolved this and is now part of the baseline configuration.

Data exfiltration is genuinely hard to separate from normal large transfers. The features that make exfiltration detectable, specifically large sustained outbound flows, overlap heavily with legitimate file sync operations. Adding destination reputation scoring and time-of-day context as features would likely improve recall on this class without significantly increasing false positives elsewhere.

IX. FUTURE WORK

Several concrete improvements follow naturally from the limitations above.

Live telemetry integration. Replacing the log export step with a stream-based pipeline using a NetFlow collector or eBPF agent would allow the model to run against live traffic. From there it becomes possible to automate responses: pushing a quarantine ACL for a flagged source IP without waiting for operator intervention, for example. Apache Kafka is a practical choice for stream ingestion and Faust for stream-based processing.

Full dual-stack IPv6 deployment. IPv6 is currently only on core inter-router links. Rolling it out to VLANs 10, 20, and 30 would make the simulation reflect what modern enterprise networks actually look like. It would also require revisiting the existing ACL and NAT configurations for IPv6 compatibility [8].

HTTP/2 and HTTP/3 support. Moving from Packet Tracer to GNS3 with Linux containers would allow deploying an HTTP/2-capable web server such as nginx and experimenting with HTTP/3 over QUIC. Measuring handshake latency differences between TLS over TCP and QUIC on the same link would be a useful addition to the application layer analysis.

Larger and more diverse training data. The model should be retrained regularly as normal traffic patterns shift. Adding labeled samples from public IDS datasets such as CICIDS-2018 and UNSW-NB15 alongside locally generated data would improve generalization to real-world conditions. Active learning could reduce the manual labeling burden by identifying the most informative samples to prioritize.

Prometheus and Grafana for observability. Replacing the custom Flask dashboard with a Prometheus and Grafana stack would give the project production-grade metrics storage, configurable alerting, and a much richer visualization layer. SNMP exporters for Cisco IOS already exist within the Prometheus ecosystem, so this is a low-effort integration with significant operational upside.

X. CONCLUSION

This project shows that meaningful network security and intelligent traffic monitoring are achievable within the constraints of a simulation environment using standard protocols and open tooling. A well-segmented VLAN topology, per-zone ACL policies, encrypted management channels, and centralized logging together handle the majority of realistic enterprise threat scenarios without needing specialized hardware or commercial software.

Adding a behavioral detection layer on top of that baseline extends the system's coverage to attacks that rule-based defenses would miss entirely. A weighted F1-score of 0.92 across five attack categories, combined with SIEM-level correlation that cross-references model output with firewall logs, produces something more useful than either component would be alone.

The TCP experiment confirms what the RFC literature predicts: Cubic outperforms Reno by a significant margin under packet loss. A 4× throughput ratio under 5% loss is a concrete number that has real implications for transport configuration decisions on campus WAN links.

The biggest remaining gap is the loose coupling between the simulation and the detection pipeline. Getting that closer to a real-time integration is the single improvement that would have the most impact on the practical usefulness of this system.

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